

Intrinsics4Tun: a proxy-based reflective framework for label-free tunnel point-cloud segmentation at deployment

Highlights

- Label-free point cloud tunnel-lining segmentation at deployment.
- Ring-level projection-based pipeline employing boundary detection.
- Gravity-referenced tunnel bottom alignment is essential.
- Offline BO-calibrated intrinsic proxy serving as an mIoU indicator.
- Self-refinement increasing mean mIoU 3.46-fold over static SAM4Tun.

Abstraction

Automated segmentation of tunnel-lining components from point clouds enables segment-wise inspection of joint gaps, leakage, and deformation. Deployment typically lacks ground-truth labels, yet the system must still adapt and decide whether to accept, refine, or flag each result. We present **Intrinsics4Tun**, a deployment-time label-free quality control framework for a projection-based, ring-level tunnel-lining segmentation pipeline. Offline, BO is used as an experience generator on a small labelled calibration panel to identify sensitive parameters and intrinsic diagnostics that correlate with mIoU; these are then frozen as an intrinsic proxy for deployment-time self-assessment and refinement. We further expose a structural failure mode of ring-level projection: unresolved rotational ambiguity can produce physically misaligned segmentations, which we address with a gravity-referenced bottom-stabilisation prior that anchors each ring before projection. During deployment, LLM-proposed/reflection-based adaptations search bounded candidate layouts using the

calibrated proxy. On 50 held-out rings from five tunnels, mean mIoU improves from 0.137 for the static SAM4Tun baseline to 0.475 after intrinsic refinement, a 3.46-fold increase. The rule-based seeded Intrinsic4Tun pipeline reaches 0.167 (+21.5%), bottom stabilisation reaches 0.290 (+73.9% over seeded), and proxy-guided refinement provides the final lift to 0.475 (+63.7% over stabilised). These results show that compact intrinsic diagnostics can support label-free adaptation and quality control for auditable tunnel-lining segmentation.

1. Introduction

Gap

At deployment, without labelled data, segmentation systems need to perform both adaptation and quality control (whether to accept, refine, or flag).

- Engineer-designed systems are auditable but less adaptable (manually tuning).
- Deep-learning systems are adaptable with a labelled dataset but less auditable.

Intrinsic4Tun is label-free at deployment and the proxy is constructed by observable internal qualities.

Intrinsics4Tun

Pipeline

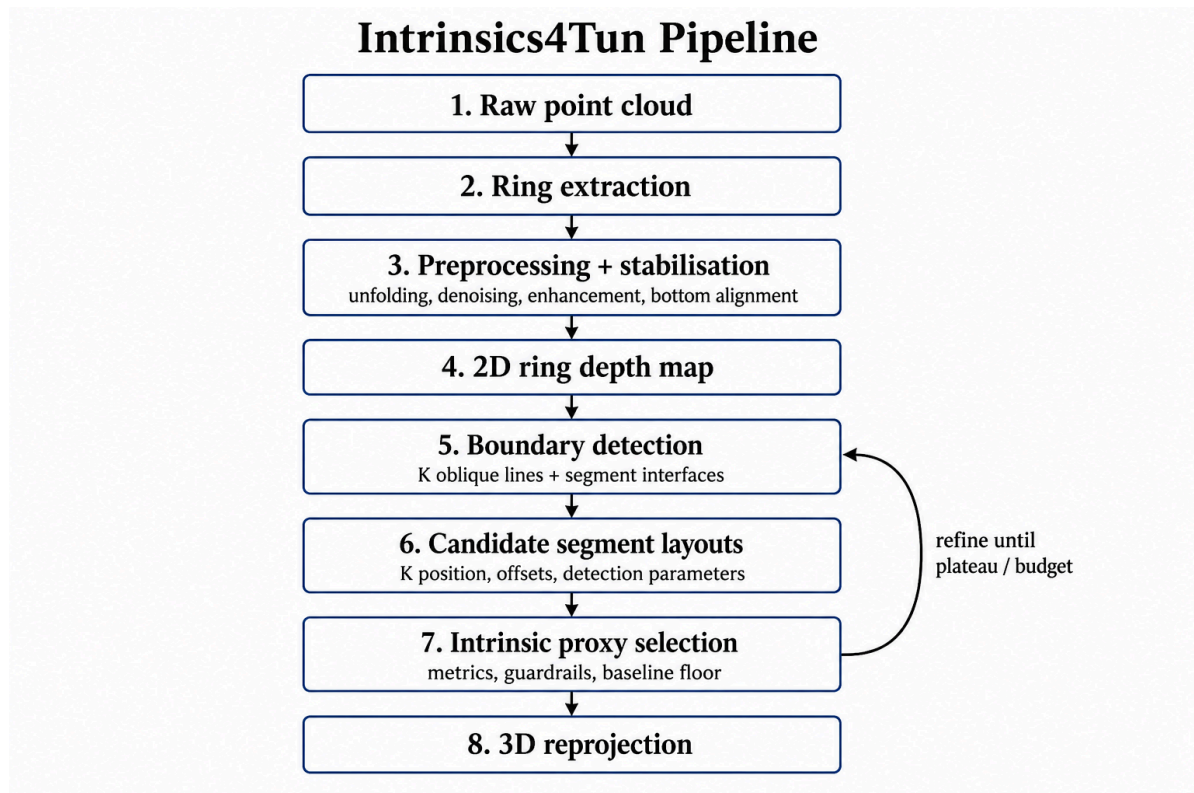


Figure 1: Intrinsics4Tun converts each ring point cloud into a stabilised 2D depth map, detects candidate segment boundaries, selects the best segment layout using an intrinsic proxy, and reprojects the selected segmentation to 3D.

- The point cloud is processed at ring level by projecting the tunnel surface into a 2D depth map through unfolding, denoising, and enhancement.
- Domain-informed structural priors are used to initialise parameters, including tunnel geometry, expected segment count and cyclic order.
- Boundary detection is then performed on the depth map, using oblique-line evidence for the keystone region and near-horizontal boundary evidence for the remaining segment interfaces.
- Candidate segment layouts are generated by varying K position, segment offsets, and related detection parameters via LLM-proposed/reflection-based adaptations, but they are accepted only when intrinsic structural metrics improve.

- The iterative process continues within a fixed budget and terminates when intrinsic improvement plateaus or the budget is exhausted.
- The selected 2D segmentation is reprojected back to the 3D point cloud; mIoU is used only for offline evaluation.

Intrinsic proxy

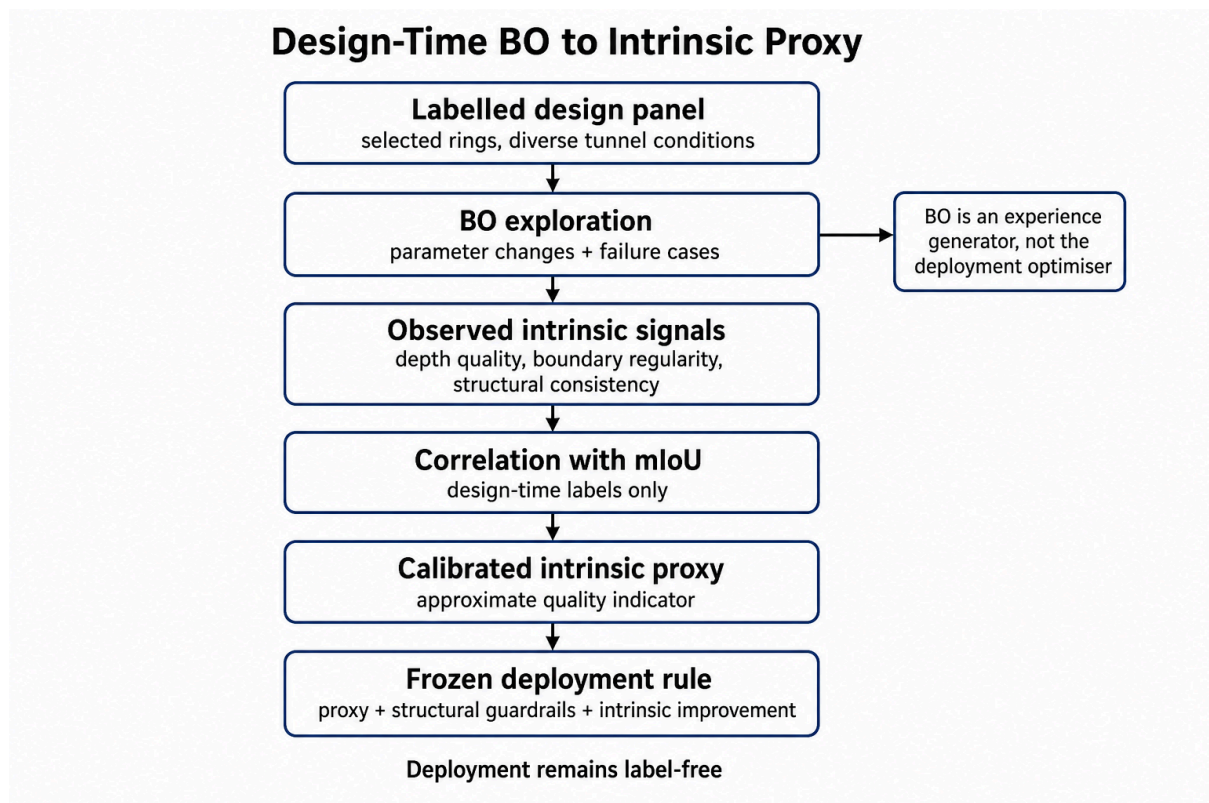


Figure 2: Design-time construction of the intrinsic proxy. BO is used on a small labelled design panel as an experience generator to expose parameter sensitivities, failure cases, and intrinsic signals that correlate with mIoU. The calibrated proxy and structural guardrails are then frozen for label-free deployment.

- It uses a small selected labelled design dataset covering diverse tunnel and point-cloud conditions to calibrate the relationship between intrinsic quality measures and segmentation mIoU.
- At design time, BO is used to identify sensitive parameters and construct an intrinsic proxy that acts as an approximate mIoU indicator.
- At deployment-time, the system remains label-free.

- It initialises the pipeline using domain-informed structural priors, including tunnel geometry, expected segment count, cyclic segment order, and gravity-referenced bottom alignment;
- Candidate adaptations, including LLM-proposed/reflection-based variants, are evaluated using the intrinsic proxy and structural guardrails;
- A candidate is accepted only when it improves the intrinsic objective and does not fall below the stabilised baseline safety floor; otherwise the system abstains to the baseline.
- Iteration terminates when intrinsic gains saturate or the adaptation budget is exhausted.

Key observations

Problem:

- During ring-level projection, rotational ambiguity can cause the unwrapped tunnel surface to start at inconsistent angular positions, leading to misaligned segment boundaries.

Solution:

- Gravity-referenced tunnel-bottom alignment resolves this ambiguity by anchoring the unfolding orientation to a physically aligned reference.



(a)



(b)

Figure 3: Example of gravity-referenced tunnel-bottom alignment on the same ring (T2-1/r60). (a) Before stabilisation, the unwrapped ring is not anchored to the tunnel bottom, so the usable lining surface is split by a large blank band in the depth map. This

white band corresponds to the non-informative gap/floor-side region introduced by the arbitrary rotational phase and cropping. (b) Gravity-referenced bottom alignment rotates the unfolding into a consistent orientation, placing the relevant lining surface into a continuous analysis window for boundary detection and segmentation.

Contributions:

1. We formulate label-free deployment quality control for tunnel-lining point-cloud segmentation using intrinsic, observable pipeline metrics.
2. We use design-time Bayesian Optimization (BO) on a small labelled calibration set to identify intrinsic metrics and parameter sensitivities that correlate with segmentation quality.
3. We identify rotational ambiguity in ring-level 3D-to-2D unfolding as a structural failure mode, and introduce gravity-referenced tunnel-bottom alignment to stabilise the projection.
4. We construct an intrinsic proxy for estimating segmentation quality, enabling accept/refine/abstain decisions without deployment labels.
5. We integrate these components into an auditable end-to-end framework, improving mean mIoU by approximately 3.46-fold over the static baseline.

2. Related work

- Tunnel-lining and underground infrastructure point-cloud segmentation
- Deployment-time segmentation quality control without labels
- BO for design-time calibration and proxy training
- LLM-guided iterative refinement / agentic optimisation.

3. Methodology

Task definition

Given a segmental tunnel-lining ring point cloud, the task is to identify the keystone segment (K), its two adjacent segments, and the remaining base segments.

- Each ring contains exactly one K segment.
- The angular position of K may vary across rings.
- The total number of lining segments is typically 6 or 7.
- Segment labels follow a fixed cyclic order, for example K, B1, A1, A2, A3, A4, B2, up to rotation
- Non-K segment offsets are expected to remain relatively consistent within the same tunnel type.

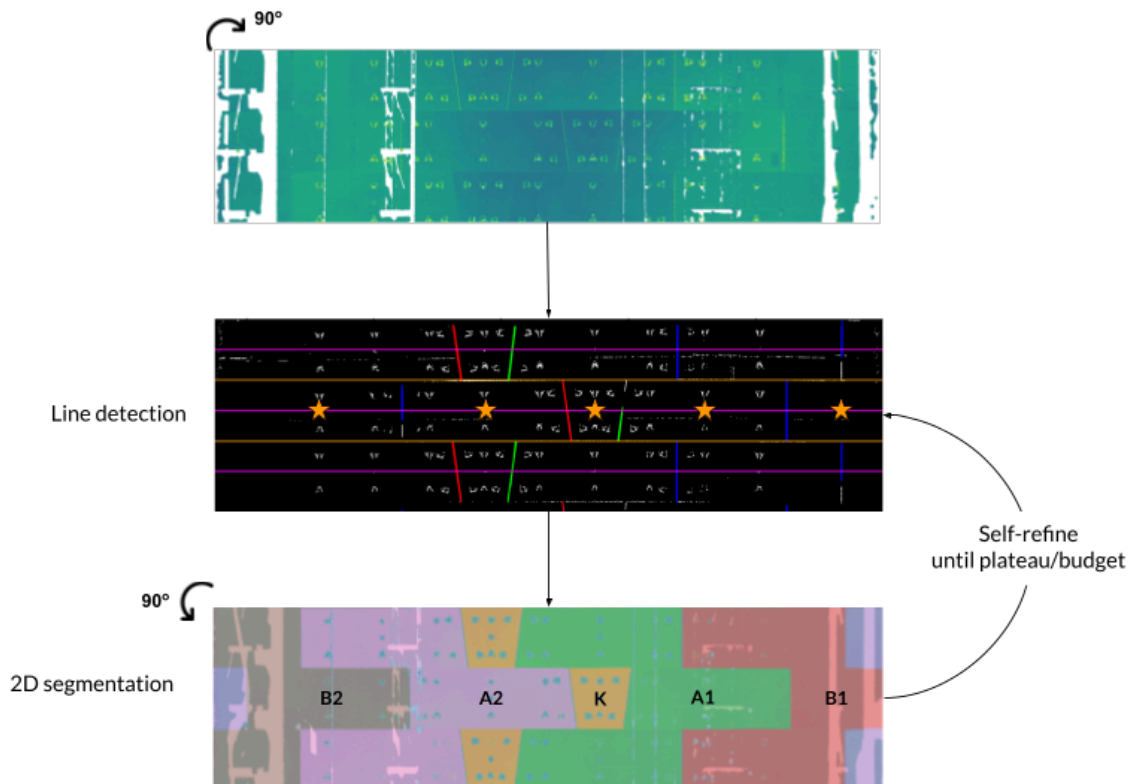


Figure 4: Example ring-level boundary detection for the segmental lining task. The detected horizontal interfaces define the ordered base segments, while the keystone region is identified from the distinctive central/oblique line evidence. The goal is to iteratively recover the single K segment, its two adjacent segments, and the remaining cyclically ordered base segments.

Dataset

- BO calibration panel:
 - 6 labelled K-bearing, covering density, coverage, K-span, and joint-orientation variations.

- Held-out evaluation panel:
 - 50 random rings, 10 each from five tunnels, including two regular staggered (T1-2), one continuous (T3), and two irregular staggered (T4-5).

Table 1. Composition of the BO calibration and held-out evaluation panels. The BO panel contains six selected labelled rings used only at design time, while the held-out panel contains 50 rings used for evaluation. Descriptors summarise point density, ring coverage, keystone span, and cyclic joint pattern diversity across the panels.

Descriptor	BO calibration	Held-out
Rings	6	50
Density (dense / medium / low+sparse)	2 / 2 / 2	13 / 18 / 19
Coverage (full / partial)	5 / 1	48 / 2
K-span (narrow / normal / wide)	1 / 3 / 2	3 / 22 / 25
Joint pattern (canonical / reversed canonical)	3 / 3	26 / 24

BO

BO is used as a design-time experience generator, not as the final optimisation engine. On the labelled calibration panel, BO explores parameter changes and failure cases to reveal which intrinsic, observable signals track mIoU/OA. These signals are then distilled into a frozen proxy and guardrails for label-free deployment.

(Potential figure exemplar illustrating how BO works)

Proxy

Employ the Ridge model for standard offline supervised, GT-based mIoU training on BO-derived data. During deployment, runtime scoring is exclusively performed using the three non-GT observable features.

(Potential figure exemplar illustrating how Ridge model works)

Ablation study

- SAM4Tun static parameters
- Rotation-ambiguous with initialised parameters as baseline
- Stabilised unfolding with gravity-referenced alignment
- Intrinsic reflection iterations within budget

Table 2. Fixed adaptation budget used during iterative intrinsic refinement. Each ring is allowed up to three refinement rounds, with at most 18 candidate layouts evaluated per round. The search terminates early after one non-improving round.

Budget item	Value
Max rounds per ring	3
Patience	1 non-improving round
Max candidates per round	18

Evaluation metrics

- mIoU/OA
- CI
- P-value

4. Results

Proxy

proxy_miou = 0.015213

-0.031164 * struct_missing_ids

$+0.196603 * \text{depth_row_nonempty_ratio}$

$+0.081910 * \text{geom_boundary_gap}$

- `struct_missing_ids`
 - *What it is:* How many expected segment IDs were missing before completion/repair.
 - *Intuition:* More missing IDs means the structure is less complete, so quality should drop.
 - *Coefficient sign:* Negative (-0.023682) -> more missing pieces lowers predicted mIoU.
- `depth_row_nonempty_ratio`
 - *What it is:* The fraction of vertical rows in the depth image that contain at least one valid depth value, computed and recorded by the pipeline's depth-quality audit file.
 - *Intuition:* Higher audited coverage usually means cleaner depth input and easier segmentation.
 - *Coefficient sign:* Positive (+0.201192) -> strongest term in this formula.
- `geom_boundary_gap`
 - *What it is:* The coefficient of variation (std/mean) of consecutive boundary spacings around the ring, derived from candidate boundary positions (after wrap-around).
 - *Intuition:* It measures how uneven the candidate's boundary layout is; this captures candidate-specific geometric structure that helps distinguish competing candidates within the same ring.
 - *Coefficient sign:* Positive (+0.081910) -> higher boundary-gap variability is associated with higher predicted mIoU.

Overall performance

Overall, Intrinsics4Tun increases mean mIoU from 0.1374 for the static SAM4Tun baseline to 0.4750 after refinement, a total lift of +0.3376 and about 3.46x relative improvement. The largest gains come from stabilisation (+0.1233) and intrinsic refinement (+0.1849), showing that both tunnel-bottom stabilisation and proxy-guided adaptation contribute materially to final performance.

Table 3. Overall segmentation performance on the 50-ring held-out panel. Mean mIoU is reported for the static SAM4Tun baseline and three Intrinsic4Tun stages. Lift denotes the incremental gain over the previous stage. Confidence intervals are 95% CIs across rings, and p-values are paired tests against the preceding stage.

Stage	mIoU value	Lift	95% CI	P-value
SAM4Tun: static	0.1374	-	[0.0719, 0.2029]	Reference
Intrinsic4Tun: seeded	0.1668	+0.0295	[0.1381, 0.1955]	0.318 vs static
Intrinsic4Tun: stabilised	0.2901	+0.1233	[0.2300, 0.3503]	1.64e-4 vs seeded
Intrinsic4Tun: refined	0.4750	+0.1849	[0.4121, 0.5379]	8.90e-4 vs stabilised

Table 4. Per-tunnel performance improvement of Intrinsic4Tun over SAM4Tun static baseline. Values report mean mIoU per tunnel. Lift is computed as final refined Intrinsic4Tun minus SAM4Tun static on paired rings. SAM4Tun fails to adapt to T3-T5 because of tunnel diameter and segment-count differences, while Intrinsic4Tun remains adaptive through stabilisation and intrinsic refinement.

Tunnel	SAM4Tun	Intrinsic4Tun	Lift	CI	P-value
T1	0.2094	0.4574	+0.2480	[0.0654, 0.4306]	0.0133
T2	0.4637	0.6566	+0.1928	[0.0022, 0.3835]	0.0479
T3	0.0000	0.5349	+0.5349	[0.4014, 0.6685]	1.5e-5
T4	0.0000	0.4187	+0.4187	[0.2178, 0.6197]	0.0011
T5	0.0000	0.3528	+0.3528	[0.1994, 0.5063]	5.6e-4

Cross-model consistency

(Not sure, we are using Claude Opus 4.7)

5. Discussion

Key findings

The results suggest three main findings.

1. **Ring-level segmentation is an effective formulation for tunnel-lining point clouds.** By decomposing the tunnel into structurally meaningful rings, the segmentation task becomes more constrained and easier to regularise using geometric priors such as segment count, cyclic order, and keystone position.
2. **A well-calibrated intrinsic proxy can provide effective label-free quality control.** Although the proxy is calibrated using labelled data at design time, deployment does not require ground-truth labels. The system can evaluate candidate segmentations using observable structural signals and select improved outputs without directly computing mIoU.
3. **Intrinsic self-refinement enables practical adaptation.** Instead of relying on a fixed parameter setting, Intrinsic4Tun can search over candidate adaptations and accept those that improve the intrinsic objective. This allows the pipeline to adapt to variations in tunnel diameter, segment count, point density, and ring geometry while remaining auditable.

Limitations

The intrinsic proxy is not perfectly accurate. It approximates segmentation quality using a small number of observable indicators, and therefore can miss some ring-level failure cases or produce regressions when the proxy score and true mIoU are not fully aligned (Table 6 in Appendix). In this study, the proxy is deliberately kept compact, using fewer than five main indicators, which improves interpretability but limits expressiveness.

The intrinsic search is also not exhaustive. The current refinement process evaluates a bounded set of candidate adaptations, so finer-grained parameter exploration may identify additional improvements. Future work could improve search efficiency by learning from accumulated self-refinement experience, for example through a reinforcement-learning (RL) policy, a retrieval-based knowledge base, or a learned proposal model that prioritises promising candidate adaptations.

6. Conclusions

This paper presents Intrinsic4Tun: a compact intrinsic proxy for deployment-time, label-free quality control in tunnel-lining point-cloud segmentation. The proxy is calibrated at design time using a small labelled panel and BO-generated experience, then fixed for deployment. At runtime, it supports accept/refine/abstain decisions using observable internal signals, without ground-truth labels. Results on the held-out panel show that proxy-guided refinement improves mIoU over static and pre-refinement stages. Gravity-referenced stabilisation is an important prerequisite because it reduces projection inconsistency and makes candidate scoring more comparable across rings. The current proxy is intentionally compact and interpretable, but this also limits representation capacity in difficult cases. Future work will focus on improving proxy calibration and ranking reliability, adding uncertainty-aware abstention, and improving candidate proposal efficiency while keeping runtime decisions auditable.

Appendix

Table 5. Ring-level mIoU progression across the 50-ring held-out panel. Each row reports one held-out ring, grouped by tunnel, with mIoU at three Intrinsic4Tun stages: seeded initialisation, gravity-stabilised preprocessing, and final intrinsic refinement. The table shows how stabilisation and proxy-guided refinement contribute differently across tunnel conditions, with the refined stage producing the final selected segmentation used for overall performance reporting.

Ring	Tunnel	Seeded	Stabilised	Refined
1-1/r18	1	0.095	0.253	0.325

1-1/r19	1	0.093	0.093	0.437
1-2/r58	1	0.104	0.236	0.314
1-2/r59	1	0.092	0.092	0.427
1-3/r125	1	0.172	0.172	0.288
1-3/r131	1	0.121	0.121	0.319
1-4/r197	1	0.227	0.227	0.800
1-4/r204	1	0.187	0.187	0.757
1-5/r270	1	0.072	0.072	0.544
1-5/r273	1	0.114	0.114	0.362
2-1/r60	2	0.076	0.076	0.674
2-1/r64	2	0.390	0.390	0.662
2-2/r141	2	0.097	0.097	0.674
2-2/r143	2	0.369	0.369	0.731
2-3/r220	2	0.072	0.072	0.649
2-3/r224	2	0.378	0.378	0.551
2-4/r298	2	0.092	0.092	0.614
2-4/r304	2	0.426	0.426	0.630
2-5/r353	2	0.377	0.377	0.715
2-5/r360	2	0.073	0.073	0.667
3-1-1/r28	3	0.127	0.127	0.564
3-1-1/r31	3	0.149	0.149	0.786
3-1-1/r32	3	0.249	0.249	0.408
3-1-1/r36	3	0.073	0.073	0.082
3-1-2/r46	3	0.181	0.181	0.595
3-1-2/r47	3	0.201	0.201	0.419
3-1-2/r48	3	0.135	0.135	0.559

3-1-3/r77	3	0.128	0.128	0.285
3-1-3/r78	3	0.119	0.119	0.790
3-1-3/r86	3	0.377	0.377	0.408
4-1/r110	4	0.076	0.629	0.183
4-10/r398	4	0.196	0.830	0.190
4-2/r142	4	0.190	0.610	0.202
4-3/r177	4	0.169	0.585	0.459
4-4/r212	4	0.075	0.159	0.815
4-5/r249	4	0.159	0.411	0.186
4-6/r276	4	0.122	0.128	0.356
4-7/r308	4	0.081	0.173	0.814
4-8/r332	4	0.097	0.599	0.198
4-9/r363	4	0.188	0.164	0.784
5-1/r118	5	0.190	0.596	0.254
5-2/r140	5	0.166	0.581	0.177
5-3/r192	5	0.123	0.609	0.258
5-3/r195	5	0.401	0.171	0.255
5-4/r227	5	0.137	0.780	0.170
5-5/r254	5	0.163	0.408	0.167
5-5/r259	5	0.162	0.169	0.833
5-6/r285	5	0.130	0.159	0.402
5-7/r317	5	0.067	0.643	0.565
5-7/r322	5	0.082	0.447	0.447

Table 6. Regression cases where intrinsic refinement underperforms stabilised preprocessing. The table lists held-out rings for which the final refined Intrinsics4Tun result has lower mIoU than the stabilised stage. These 12 cases, all from T4 and T5, indicate proxy-selection failures

on irregular tunnel conditions where the intrinsic score favoured a candidate that was structurally plausible but worse under ground-truth mIoU.

Ring	Tunnel	Stabilised	Refined	Drop
4-10/r398	4	0.830	0.190	-0.640
4-1/r110	4	0.629	0.183	-0.446
4-2/r142	4	0.610	0.202	-0.408
4-8/r332	4	0.599	0.198	-0.401
4-5/r249	4	0.411	0.186	-0.225
4-3/r177	4	0.585	0.459	-0.126
5-4/r227	5	0.780	0.170	-0.610
5-2/r140	5	0.581	0.177	-0.404
5-3/r192	5	0.609	0.258	-0.351
5-1/r118	5	0.596	0.254	-0.342
5-5/r254	5	0.408	0.167	-0.241
5-7/r317	5	0.643	0.565	-0.079